

MamNet-6: Multi-View Fusion, CrossView Transformer, and Uncertainty-Aware Triage for Breast Cancer Detection on VinDr-Mammo

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Extended Abstract—Breast cancer is the most commonly diagnosed cancer worldwide. Population screening with full-field digital mammography (FFDM) reduces mortality but places a heavy radiologist workload, with inter-reader variability of 10–30 percentage points between clinicians (McKinney et al., 2020). Deep learning has shown strong promise—meta-analyses report pooled AUC of 0.87 for AI versus 0.81 for radiologists on digital mammography (Yoon et al., 2023)—yet most published work on the VinDr-Mammo benchmark processes one view at a time, ignoring the CC/MLO pairing that radiologists rely on. We address this by building MamNet-6, a six-stage pipeline that progressively adds multi-view fusion, spatial cross-attention, uncertainty quantification, and hybrid triage simulation.

We evaluated MamNet-6 on VinDr-Mammo (Nguyen et al., 2023), a 20,000-image Vietnamese FFDM benchmark. Starting from single-view CNN baselines across five backbones (Stage 1), we introduced BreastPairNet—a shared-backbone network that jointly processes CC and MLO views across twelve backbone×fusion×resolution variants (Stage 2)—then added a CrossView Transformer with spatial cross-attention between view feature maps (Stage 3). A parallel ablation tested whether adding coarse bounding-box annotations improves results (Stage 3b); it did not. Stage 4 applies Monte Carlo Dropout ($N=20$ passes) to produce calibrated uncertainty estimates, and Stage 5 assembles a 16-model ensemble with test-time augmentation. In Stage 6, we simulate a Verboom-style triage policy (Verboom et al., 2025) where the AI handles its most confident reads autonomously and defers the rest to a radiologist. We validate zero-shot on the RSNA 2022 dataset.

DenseNet-121 achieved AUROC = 0.756 as a single-view baseline. BreastPairNet brought this to 0.826 [95 % CI 0.776–0.871] (+9.1 %) and the CrossView Transformer to 0.832 [0.783–0.875] (+10.0 %); individual stage gains sit within overlapping confidence intervals, so the cumulative Stage 1→5 improvement is the more reliable comparison. Bounding-box annotations degraded AUROC by 0.12–0.20 across all variants—a consistent shortcut effect we did not anticipate—suggesting the model learns to exploit box-region features rather than the underlying pathology. The 16-model ensemble achieved AUROC = 0.846 (+11.8 %) with sensitivity at 90 % specificity ($S@90\%spec$) = 0.657 and PPV = 0.184 at the triage threshold. Hybrid triage at 50 % AI reads preserved 87.9 % sensitivity under oracle conditions. Zero-shot transfer to RSNA 2022 yielded AUROC = 0.601, a substantial drop that points to a real domain gap between Vietnamese and North American screening populations.

The clearest result is that multi-view fusion, cross-attention, and ensembling compound reliably—an 11.8 % cumulative AUROC gain over single-view CNN—while each individual step is modest and within bootstrap noise. Two unexpected findings stand out: coarse localisation annotations hurt rather than help, and the RSNA generalisation gap remains wide, suggesting the model has overfit to Vietnamese scanner characteristics and does not transfer cleanly to North American screening data.

References

- McKinney, S. M. et al. (2020). International evaluation of an AI system for breast cancer screening. *Nature*, 577, 89–94.
- Nguyen, H. T. et al. (2023). VinDr-Mammo: A large-scale benchmark dataset for computer-aided diagnosis in full-field digital mammography. *Scientific Data*, 10, 277.
- Verboom, S. D. et al. (2025). AI Should Read Mammograms Only When Confident. *Radiology*, 316(2), e242594.
- Yoon, J. H. et al. (2023). Standalone AI for breast cancer detection at screening digital mammography and digital breast tomosynthesis: A systematic review and meta-analysis. *Radiology*, 307(5), e222639.